

Acid–Base Properties of Salts

Hydrolysis, $K_a \times K_b = K_w$ and salt pH

Name _____ Class _____ Date _____

DATA AT 25 °C

$$K_w = 1.0 \times 10^{-14} \cdot K_a \times K_b = K_w \cdot \text{pH} + \text{pOH} = 14.00$$

$$K_a: \text{CH}_3\text{COOH } 1.8 \times 10^{-5} \cdot \text{HF } 6.8 \times 10^{-4} \cdot \text{HCN } 4.9 \times 10^{-10} \cdot K_b: \text{NH}_3 1.8 \times 10^{-5}$$

For a weak acid or base: $x \approx \sqrt{(K \times C)}$, if x is under 5 % of C

Write the hydrolysis equation for the ion that reacts. Check the 5 % rule.

A. Predicting

1. Predict whether each solution is acidic, basic or neutral, and name the ion responsible: (a) NaNO_3 (b) K_2CO_3 (c) NH_4Br (d) CH_3COOK (e) LiCl (f) NH_4NO_3

2. Write the equation for the reaction with water of (a) the fluoride ion (b) the ammonium ion.

B. K_a and K_b

3. Calculate: (a) K_b of F^- (b) K_a of NH_4^+ (c) K_b of CN^- . Which is the strongest base, F^- or CN^- ?

C. pH of salt solutions

4. Calculate the pH of 0.10 mol/L NaCN.

5. Calculate the pH of 0.050 mol/L NH_4NO_3 .

6. Calculate the pH of 0.25 mol/L potassium ethanoate, CH_3COOK .

D. Titrations

7. 25.00 mL of 0.100 mol/L ethanoic acid is titrated with 0.100 mol/L NaOH. (a) What volume of NaOH reaches the equivalence point? (b) What is the concentration of ethanoate at that point? (c) Calculate the pH at the equivalence point and suggest a suitable indicator.

E. Challenge

8. Explain why a solution of aluminium chloride, AlCl_3 , is acidic even though Cl^- is neutral.